
RL2Ocean Calibration and Validation ATBD

Release 0.2

RL2Ocean consortium

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Table 1: Document information

Document ID	ESA-RL2OCEAN-CALVAL-ATBD
Issue	0.2
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Status	Draft (approval status not stated in source)
Processor	ROSE-L L2 Ocean Processor (RL2Ocean)
Copyright	RL2Ocean consortium
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RL2OCEAN

¹ <https://creativecommons.org/licenses/by/4.0>



CHAPTER**ONE**

AUTHORS AND APPROVAL

The LaTeX cover lists author placeholders rather than individual names.

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Author 4 (name not provided)	Delft University of Technology (TU Delft)
Author 5 (name not provided)	German Aerospace Center (DLR)

Approval and authorisation records are not specified in the source.

**CHAPTER
TWO**

CHANGE RECORDS

Issue	Date	Description	Author
0.2	Not stated	Source issue	RL2Ocean consortium
0.2	2026-09-29	Sphinx migration	RL2Ocean consortium

Note

The source abstract contains only Lorem ipsum placeholder text and is not reproduced. The source's substantive content and section outline are preserved; headings without body text are identified in the relevant chapters rather than filled with invented material.

2.1 Introduction

2.1.1 Purpose and scope

The Copernicus Programme, established by the European Union, is a framework for Earth Observation and Monitoring, designed to support environmental protection, civil protection, and civil security efforts. At the heart of this programme is the Copernicus Space Component (CSC), which comprises the Sentinel missions. These missions are integral to providing essential data across various thematic domains, thereby facilitating informed decision-making and policy development.

One of the new advancements within this framework is the Copernicus Radar Observing System for Europe – L-Band (ROSE-L). This initiative is aimed at enhancing existing capabilities by introducing L-band Synthetic Aperture Radar (SAR) data. The L-band SAR is set to complement the C-band radar data provided by Sentinel-1, thus addressing specific observational needs within environmental and maritime surveillance.

2.1.2 Cal/Val context

Quality Assessment (QA) is required for remote sensing and Earth Observation (EO) instruments in order to produce reliable measurements and ensure accuracy. Quality of products are commonly assessed through a Calibration and Validation process (Cal/Val). Consequently, the development of the prototype and operational ROSE-L Ocean products processors (L2O-GPP and L2O-IPF respectively) must be followed by the validation of products. A validation strategy is meant to be planned prior the calibration and validation activities. The required strategy shall cover different stages of the processor and different phases of ROSE-L mission, from pre-launch (phase D) to exploitation (phase E2). The strategy shall include methodologies, datasets, performance and uncertainty metrics.

The ROSE-L instrument will operate in scanning synthetic aperture radar (ScanSAR) mode. The system will support Dual-Pol, Quad-Pol and Wave modes. The Cal/Val strategy is also meant to cover all modes and polarizations. ROSE-L will not be the first SAR system functioning within L-band. ALOS, ALOS-2, ALOS-4, SAOCOM and NiSAR are, or have been, operating in L-band too. However, there are differences between those systems and the ROSE-L system, e.g. incidence angle, range bandwidth, swath width, among others. Furthermore, ROSE-L incorporates the use of a new technique, the Scan-On-Receive (SCORE) technique with Multiple Azimuth Phase



Centers (MAPS). The different instrument design will impose differences in the retrievals at low level which are meant to influence the inversion into L2 products.

Regarding the development of a retrieval algorithm but also Cal/Val activities, instrument differences between ROSE-L and previous SAR L-band instruments impose a challenge; the need of producing and validating ocean products from ROSE-L scenes during the pre-launch phase. The lack of real ROSE-L data will be mitigated by the use of similar acquisitions by the above-mentioned systems; although this introduces a non-negligible degree of uncertainty. For Cal/Val purposes, simulated ROSE-L scenes will be produced. The achievable accuracy of simulated ROSE-L will also be analysed during the In-Orbit-Commissioning (IOC) phase, E1. Once real ROSE-L data is available, the outputs from the L2O-IPF processor will need to be validated again, with the possibility to encounter a need for a readjustment of the processor or introduction of supplementary calibration steps.

A tool for assessing the quality of the ocean products is vital. The Cal/Val tool, L2O-QC, is meant to be developed in parallel to the development of the ocean processors, L2O-GPP and L2O-IPF. The L2O-QC software is required to be able to ingest datasets of different nature; ROSE-L acquisitions, reference datasets (either in-situ or remote), and auxiliary data. The goal is to collocate the different datasets in order to undertake the quality assessment, where the accuracy and reliability of the ROSE-L L2 ocean products will be determined and ensured. The underlying objective is to provide users with a set of performance and uncertainty metrics necessary for a deep understanding of the products and posterior correct usage and application.

2.1.3 L2 product scope

The L2 products to validate are already listed in the objectives of L2O-GPP and L2O-IPF: Ocean Surface Vector Stress (OSVS), Sea State, and line-of-sight (LoS) surface currents. Within each of the three thematic products, the following L2 variables are meant to be produced and therefore validated:

- OSVS:
 - Wind vector
 - Wind stress
 - Hub height wind speed
- Sea state:
 - Total significant wave height
 - Mean period wind-sea
 - First moment wave period
 - Second moment wave period
 - Wave height swell dominant system
 - Wave height swell secondary system
 - Significant wave height wind-sea
 - Mean period wind-sea
 - Wave direction
 - Wave directional energy spectrum
- LoS currents:
 - LoS wave Doppler
 - LoS surface current component

2.1.4 Cal/Val objectives and performance

The key objectives of the Calibration and Validation (Cal/Val) phase include ensuring that all L2 geophysical products meet internationally recognized standards, such as SI traceability, and align with QA4EO principles to guarantee high data quality. Cal/Val aims to validate the accuracy, precision, and stability of L2 products using independent datasets, such as Fiducial Reference Measurements (FRM), and cross-sensor harmonization. A critical

focus is on quantifying and documenting uncertainty propagation from instrument measurements to retrievals while ensuring the retrieval algorithms are robust under diverse environmental conditions.

To meet performance requirements, L2 products must achieve predefined thresholds in accuracy, precision, and stability, with uncertainties traceable and reliable across pixel and aggregate levels. Products must demonstrate consistency over full mission coverage, addressing global, regional, and temporal scales, and maintain defined operational latencies (e.g., near-real-time or reprocessed outputs). Validation activities use measurable metrics, including statistical bias, RMS error, and spatial-temporal coverage rates, to assess product quality systematically. Lastly, the outputs should be delivered in user-ready formats with comprehensive documentation while maintaining version control and data provenance.

2.1.5 ATBD purpose

The aim of the Calibration and Validation Algorithm Theoretical Basis Document (ATBD) is to provide a comprehensive and transparent description of the scientific principles, mathematical methods, and processing steps used to generate ROSE-L L2 ocean products. It ensures that the algorithms are well-documented, reproducible, and traceable, facilitating a clear understanding of how measurements are transformed into geophysical quantities. The ATBD serves as a foundation for ensuring data quality, guiding calibration/validation efforts, and supporting user confidence in the satellite mission's outputs.

2.1.6 Applicable and reference documents

The source contains no separate applicable-document list. References cited in the uncertainty-analysis material are listed in that chapter, and the complete source BibTeX database is included as `bibliography.bib`.

2.1.7 Acronyms

CANVAS

CALibration aNd VALidation for Sar.

CEOS

Committee on Earth Observation Satellites.

DC

Doppler Centroid.

DCA

Doppler Centroid Anomaly.

DORIS

Doppler Orbitography and Radiopositioning Integrated by Satellite.

EO

Earth observation.

ESA

European Space Agency.

EU

European Union.

FDR

Fundamental Data Record.

FIDUCEO

Fidelity and Uncertainty in Climate Data Records from Earth Observation.

FP7

Seventh Framework Programme.

FRM

Fiducial Reference Measurement.

GEO

Group on Earth Observations.



GEOSS

Global Earth Observation System of Systems.

GUM

Guide to the expression of Uncertainty in Measurement.

JCGM

Joint Committee for Guides in Metrology.

QA4EO

Quality Assurance Framework for Earth Observation.

SAR

Synthetic Aperture Radar.

SI

International System of Units.

TDP

Thematic Data Product.

UTD

Uncertainty Tree Diagram.

VIM

International Vocabulary of Metrology.

2.1.8 Definitions and conventions

Accuracy

The quality of being near to the true value. The source marks this definition as requiring a reference.

Comparison

Metrologists validate uncertainty analysis and confirm traceability through comparisons. The Mutual Recognition Arrangement requires regular, formal international comparisons to establish the degree of equivalence between SI realisations and disseminations in different countries.

Dataset

A combination of data records, such as numerical values needed to analyse and interpret a natural process, and the related information content.

Fiducial Reference Measurement

A suite of independent, fully characterised, and traceable sub-orbital measurements, relevant to space-based observations, following the guidelines of the GEO/CEOS Quality Assurance Framework for Earth Observation (QA4EO).

Fundamental Data Record

A record, of sufficient duration for its application, of uncertainty-quantified sensor observations calibrated to physical units and located in time and space, together with ancillary and lower-level instrument data used to calibrate and locate the observations and to estimate uncertainty (*Woolliams et al. 2025a*).

Interoperability

The ability of data or tools from non-cooperating resources to integrate or work together with minimal effort (*Wilkinson et al. (2016)²*).

Interoperable

Describes data or tools from non-cooperating resources that can integrate or work together with minimal effort (*Wilkinson et al. (2016)³*).

Measurand

The quantity intended to be measured, and the single output quantity from a measurement model.

Metrology

The science of measurement and its application (*BIPM et al., n.d.-b*).

² <https://doi.org/10.1038/sdata.2016.18>

³ <https://doi.org/10.1038/sdata.2016.18>

Thematic Data Product

A record, of sufficient duration for its application, of uncertainty-quantified retrieved values of a geophysical variable, along with ancillary data used in retrieval and uncertainty estimation.

Traceability

A property of a measurement relating its value to a stated metrological reference through an unbroken chain of calibrations or comparisons, with uncertainties evaluated and methodologies documented at each step.

Uncertainty analysis

The review of uncertainty sources and their propagation through the traceability chain, based on the suite of documents prepared by the JCGM and known as the GUM.

2.2 Processing model and data flow

2.2.1 Overview and top-down decomposition

The source describes the calibration and validation activities across the ROSE-L mission phases. It does not provide a detailed processing-block decomposition or an algorithm-level data-flow diagram.

2.2.2 Pre-launch phase (D)

During the pre-launch phase (phase D), activities focus on preparing and validating the L2 processing chain, ensuring readiness for on-orbit operations. Key tasks include finalizing L2 requirements, algorithms, and performance budgets. Simulators and proxy datasets are leveraged to validate retrieval algorithms and uncertainty models. Pre-launch instrument characterization is integrated into the processing chain to ensure retrievals are robust to on-orbit conditions. A comprehensive Cal/Val Plan is developed, identifying validation sites, protocols, data matchups, and metrics.

2.2.3 Commissioning phase (IOC or E1)

In the commissioning phase (IOC or phase E1), the primary goal is to validate L2 products using on-orbit data and release initial beta/provisional products with documented performance. Early activities include confirming L1 performance, performing vicarious calibrations, and matching L2 products with ground-based and airborne reference datasets (FRM sites, buoys, ship-based campaigns, etc.). Algorithms are fine-tuned using real-time data, and initial uncertainty models are validated. Early reprocessing cycles are conducted to improve product stability, and performance against key metrics is assessed. Products are released with quality summaries, and ongoing monitoring dashboards are set up.

2.2.4 Exploitation phase (E2)

The exploitation phase (phase E2) ensures L2 products maintain accuracy, precision, and stability over the mission lifecycle. Routine validation is conducted using reference networks, continuous KPI monitoring, and periodic reprocessing cycles for algorithm and calibration improvements. Traceability to standards (QA4EO, FRM) and uncertainty validation is maintained, while long-term drift corrections support stable time series. Community engagement, user feedback, and updates to documentation (validation reports, product guides) are prioritised. The goal is to achieve validated product maturity, ensuring high-quality data for science and operational applications.

2.2.5 End-to-end data flow

No end-to-end processing-flow diagram, processing-step mapping, or algorithm identifiers are provided in the source LaTeX document.

2.3 Algorithms description

The source groups its method outlines into calibration and validation. The method-specific headings are retained in the following pages. Their details are not present in the source.

2.3.1 Selected Calibration methods

The source provides the following section headings but no method descriptions, input specifications, processing details, performance metrics, or expected outputs.

2.3.1.1 ROSE-L L1 SLC products

Short description

Description of inputs

Detailed methodology

Performance metrics

Expected output

2.3.1.2 NOC

Short description

Description of inputs

Detailed methodology

Performance metrics

Expected output

2.3.1.3 GMF tuning

Short description

Description of inputs

Detailed methodology

Performance metrics

Expected output

2.3.1.4 MTF tuning

Short description

Description of inputs

Detailed methodology

Performance metrics

Expected output

2.3.1.5 Summary

The source contains this heading without summary text.

2.3.2 Selected Validation methods

The source provides the following section headings but no method descriptions, input specifications, processing details, performance metrics, or expected outputs.

2.3.2.1 One-to-one validation

Short description

Description of inputs

Detailed methodology

Performance metrics

Expected output

2.3.2.2 Triple collocation

Short description

Description of inputs

Detailed methodology

Performance metrics

Expected output

2.3.2.3 Spatial variances

Short description

Description of inputs

Detailed methodology

Performance metrics

Expected output

2.3.2.4 Summary

The source contains this heading without summary text.

2.4 Uncertainty analysis

2.4.1 Introduction

The “Evaluation of measurement data – GUM” (*BIPM et al., n.d.-a*), or the “GUM bible”, is a foundational international document that provides a conceptual and mathematical framework for identifying sources of uncertainty, quantifying them in a consistent way (as standard deviations), combining them into a single uncertainty figure, and reporting the result and its uncertainty clearly and comparably across disciplines and countries. Focusing on evaluation and expression of uncertainty in measurement it is an important component of the science of measurement and application, known as metrology (e.g., *BIPM et al., n.d.-b*).

The GEO is an international partnership of governments and organizations acting as a policy and coordination body to promote global sharing and use of EO data, for example to support work on climate, disaster management, agriculture, water, and ecosystems. The GEOSS is the global, distributed infrastructure that GEO is building and coordinating. Rather than being a single system, it connects many independent EO systems – such as satellites, ground networks, and ocean buoys – from different countries and agencies, and aims to make them interoperable, accessible through common standards and interfaces, and accompanied by information about data quality and uncertainty. As such, GEO is the partnership, and GEOSS is the “system of systems” that GEO oversees.

To support its mission, GEOSS needs a metrological framework. As such, the 2010 endorsement of the QA4EO by the CEOS (part of GEOSS), has defined guidelines for EO applications based on principles adapted from the metrology community (*Woolliams et al. 2025a*). QA4EO aims to ensure that EO data are accessible, of known and documented quality, traceable to reference standards, in particular the SI, and suitable for user application (“fit

for purpose”).

This section applies the GUM principles to SAR data and reviews the existing frameworks supporting metrology: the VIM (*BIPM et al., n.d.-b*), the GUM (*BIPM et al., n.d.-a*), and QA4EO. The LaTeX source also refers to section labels for level-1, level-2, error propagation, and conclusions that are not defined there. Its product-specific uncertainty-estimation headings are retained below.

2.4.2 The GUM Principles: A review

2.4.2.1 The International Vocabulary of Metrology (VIM)

The International Vocabulary of Metrology (VIM; *BIPM et al., n.d.-b*) defines and organizes key concepts in metrology to ensure global consistency in measurement science. It provides standardized terminology covering quantities, units, and measuring instruments, with a focus on measurement uncertainty rather than error.

The document explains critical properties of measuring devices, such as precision, stability, and sensitivity, and highlights the importance of traceability and standards for reliability.

Its applications span various fields, including physics, chemistry, biology, medicine, and emerging areas like forensic science and food technology. Harmonization of terms is emphasized to ensure coherence in the various scientific, engineering, regulatory, and quality assurance fields.

2.4.2.2 Guide to the Expression of Uncertainty in Measurement (GUM)

The GUM establishes general rules for evaluating and expressing uncertainty in measurement that are intended to be applicable to a broad spectrum of measurements (*BIPM et al., n.d.-a*). The guide was developed to solve the challenge of different laboratories and disciplines using different methods and terminology for uncertainty, which made measurement results difficult to compare.

The central idea of the GUM is to treat all components of uncertainty in the same mathematical way, regardless of whether they arise from random variation or systematic effects. Every component is represented by a probability distribution and a standard uncertainty. To obtain standard uncertainties, the GUM distinguishes between two kinds of evaluation. “Type A” evaluations use statistical analysis of repeated measurements, whereas “Type B” evaluations use all other available information, such as instrument specifications, calibration certificates, or expert judgment. The results of both evaluation types can be converted into standard uncertainties so they can be used on the same terms.

Measurements are described by a mathematical model that links the quantity being measured (the measurand) to various input, or influence, quantities. Once standard uncertainties of the influence quantities are estimated, they are combined – taking into account any correlations – using standard rules for propagating variances. This yields the combined standard uncertainty of the measurand, taken as the estimated standard deviation associated with the result. An expanded uncertainty is obtained by multiplying the standard uncertainty by a coverage factor k . The purpose of the expanded uncertainty is to provide an interval about the result of a measurement that may be expected to encompass a large fraction of the distribution of values that could reasonably be attributed to the measurand. The value of k and how it was chosen must always be stated.

A key emphasis of the GUM is transparency. An uncertainty statement should clearly describe which inputs were considered, how each uncertainty component was obtained, what assumptions were made, and how the components were combined. The document supplements this framework with definitions, statistical background, guidance on practical evaluation, and worked examples. In doing so, it provides a common language and method that allow measurement results and their uncertainties to be understood and compared across borders and domains.

2.4.2.3 Quality Assurance Framework for Earth Observation (QA4EO)

GEOSS depends on interoperable and quality-assured data to derive comprehensive information about climate and environment, as well as, e.g., food, water and energy security. To achieve interoperability across systems and datasets, common and well-defined data and metadata formats are needed, but also metrological principles such as traceability to SI, documented uncertainty analysis and comparison between different measurement references must be followed, and are key to the QA4EO reports (*Woolliams et al. 2025a*).

Beside *BIPM et al., n.d.-b* and *BIPM et al., n.d.-a*, QA4EO also builds on EO-metrology collaboration in the EU FP7 and Horizon 2020, ESA and Copernicus projects and, in particular, the FIDUCEO project (*Mittaz et al. 2019*).

As summarized in (Woolliams *et al.* 2025b), three basic metrology principles generally apply to EO data:

1. **Traceability:** Metrological traceability is a property of a measurement that relates the measured value to a stated metrological reference through an unbroken chain of calibrations or comparisons. It requires, for each step in the traceability chain, that uncertainties are evaluated and that methodologies are documented.
2. **Uncertainty Analysis:** Uncertainty analysis is the review of all sources of uncertainty and the propagation of that uncertainty through the traceability chain. It is based on the suite of documents prepared by the JCGM and together known as the GUM.
3. **Comparison:** Metrologists validate uncertainty analysis and confirm traceability through comparisons. The Mutual Recognition Arrangement (a formal arrangement to confirm the degree-of-equivalence between SI realisations and disseminations in different countries) requires regular, formal international comparisons that are conducted under strict rules.

The uncertainty analysis, as outlined in the GUM, should follow these principles.

ESA has initially applied the terms FRM, FDR and TDP to describe metrologically rigorous observations of specific relevance to space-based observations (Woolliams *et al.* 2025b). The terms are increasingly being used by the broader EO community, although they have not yet been formally endorsed. The terms are defined in Woolliams *et al.* 2025b, as follows:

- **Fundamental Data Record (FDR):** An FDR is a record, of sufficient duration for its application, of uncertainty-quantified sensor observations calibrated to physical units and located in time and space, together with all ancillary and lower-level instrument data used to calibrate and locate the observations and to estimate uncertainty.
- **Thematic Data Product (TDP):** A TDP is a record, of sufficient duration for its application, of uncertainty-quantified retrieved values of a geophysical variable, along with all ancillary data used in retrieval and uncertainty estimation.
- **Fiducial Reference Measurement (FRM):** An FRM is a suite of independent, fully characterised, and traceable (to a community agreed reference, ideally SI) measurements of a satellite relevant measurand, tailored specifically to address the calibration/validation needs of a class of satellite borne sensors and that follow the guidelines outlined by the GEO/CEOS QA4EO.

As highlighted by Woolliams *et al.* 2025b, “TDPs provide higher level products that have been processed from FDRs, through algorithms which also often combine information from other FDRs (e.g., from other satellite sensors) or from external information (such as reanalysis models and/or certain non-satellite data), along with such additional information”. Guidelines for assessing observational systems with respect to FRMs have been established by CEOS.

In order to establish a systematic process for generating uncertainty budgets, and ensuring that EO data are interoperable, temporally stable, and fit for scientific, operational and decision-making purposes, the QA4EO and EO projects acknowledged in Woolliams *et al.* 2025c have adapted the stages, suggested in GUM, required to support the development of a measurement model and from that an uncertainty analysis. These EO specific stages of uncertainty analysis are given by Woolliams *et al.* 2025c as follows:

- Step 1: Define the measurand and measurement model.
- Step 2: Establish the traceability with a diagram.
- Step 3: Evaluate each source of uncertainty and fill out an effects table.
- Step 4: Calculate the data product and uncertainties.
- Step 5: Record information about the uncertainty analysis for long term data preservation purposes and summarise for today’s users.

These steps are discussed in detail in Woolliams *et al.* 2025c. The product-specific headings below retain the source’s intended level-1 and level-2 uncertainty-estimation scope.

2.4.3 Product-specific uncertainty estimation

The source provides headings for the following product analyses, but no detailed measurement models, traceability matrices, effect evaluations, uncertainty calculations, or output documentation beneath them.

2.4.3.1 Azimuth wavelength cutoff

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

2.4.3.2 MACS

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

2.4.3.3 iMACS

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

2.4.3.4 DCA

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

2.4.3.5 Texture-based azimuth cutoff

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

2.4.3.6 RVL

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

2.4.3.7 Inverse Wave Age

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

2.4.3.8 Wind products

Wind speed, wind stress, and hub height wind speed

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

2.4.3.9 Sea-state products

Total significant wave height; mean period wind-sea; first and second moment wave period; wave height for the swell-dominant and swell-secondary systems; significant wave height wind-sea; wave direction; and wave directional energy spectrum.

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

2.4.3.10 Line-of-sight current products

Line-of-sight wave Doppler and line-of-sight surface current component

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

2.4.3.11 Summary

The source contains this heading without summary text.

2.4.4 References

BIPM, IEC, IFCC, et al. n.d.-a. *Evaluation of Measurement Data — Guide to the Expression of Uncertainty in Measurement*. Joint Committee for Guides in Metrology, JCGM 100:2008. doi:10.59161/JCGM100-2008E⁴.

BIPM, IEC, IFCC, et al. n.d.-b. *International Vocabulary of Metrology - Basic and General Concepts and Associated Terms (VIM)*. Joint Committee for Guides in Metrology, JCGM 200:2012(E/F).

Mittaz, Jonathan, Christopher J. Merchant, and Emma R. Woolliams. 2019. “Applying Principles of Metrology to Historical Earth Observations from Satellites.” *Metrologia* 56 (3). doi:10.1088/1681-7575/ab1705⁵.

Woolliams, Emma, Sajedah Behnia, Agnieszka Bialek, et al. 2025a. *General Guidance on a Metrological Approach to Fundamental Data Records (FDR), Thematic Data Products (TDP) and Fiducial Reference Measurements (FRM) - Executive Summary*. National Physical Laboratory.

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Woolliams, Emma, Sajedah Behnia, Agnieszka Bialek, et al. 2025c. *General Guidance on a Metrological Approach to Fundamental Data Records (FDR), Thematic Data Products (TDP) and Fiducial Reference Measurements (FRM) - Uncertainty Analysis Process*. National Physical Laboratory.

⁴ <https://doi.org/10.59161/JCGM100-2008E>

⁵ <https://doi.org/10.1088/1681-7575/ab1705>

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